

Table 1: Accuracy (%) comparison with the state-of-the-arts on CIFAR-10, CIFAR-100 and ImageNet. †: the results implemented by author-provided code. -: no results are reported. *nwna* indicates the weights and activations are quantized to *n* bit. The best results are reported with **boldface**.

Dataset	Model (Full precision)	Bit width	ZAQ (CVPR 2021)	IntraQ (CVPR 2022)	ARC+AIT (CVPR 2022)	GDFQ (ECCV 2020)	ARC (IJCAI 2021)	Qimera (NeurIPS 2021)	AdaSG (Ours)
CIFAR-10	ResNet-20 (93.89)	3w3a	-	77.07	-	75.11 [†]	-	74.43 [†]	84.14
		4w4a	92.13	91.49	90.49	90.11	88.55	91.26	92.10
		5w5a	93.36	-	92.98	93.38	92.88	93.46	93.76
CIFAR-100	ResNet-20 (70.33)	3w3a	-	48.25	-	47.61 [†]	-	46.13 [†]	52.76
		4w4a	60.42	64.98	61.05	63.75	62.76	65.10	66.42
		5w5a	68.70	-	68.40	67.52	68.40	69.02	69.42
ImageNet	ResNet-18 (71.47)	3w3a	-	-	-	20.23 [†]	23.37	1.17 [†]	37.04
		4w4a	52.64	66.47	65.73	60.60	61.32	63.84	66.50
		5w5a	64.54	69.94	70.28	68.49	68.88	69.29	70.29
	MobileNetV2 (73.03)	3w3a	-	-	-	1.46 [†]	14.30	-	26.90
		4w4a	0.10	65.10	66.47	59.43	60.13	61.62	65.15
		5w5a	62.35	71.28	71.96	68.11	68.40	70.45	71.61
	ResNet-50 (77.73)	3w3a	-	-	-	0.31 [†]	1.63	-	16.98
		4w4a	53.02	-	68.27	54.16	64.37	66.25	68.58
		5w5a	73.38	-	76.00	71.63	74.13	75.32	76.03

Table 2: Ablation study about varied components of AdaSG. *nwna* indicates the weights and activations are quantized to *n* bit. The best results are reported with **boldface**.

Model (Full precision)	$\mathcal{L}_{ds}$	$\mathcal{L}_{as}$	$\mathcal{L}_b$	$\mathcal{L}_{BNS}$	Bit width	
					3w3a	5w5a
ResNet-18 (71.47)		✓	✓	✓	14.86	69.15
	✓		✓	✓	26.41	69.81
	✓	✓		✓	32.75	70.13
		✓	✓	✓	15.01	70.06
	✓	✓	✓		20.98	67.35
	✓	✓	✓	✓	37.04	70.29

degradation (at most 22.18% and 1.14%), implying the intuition of the cooperation between $\mathcal{L}_{ds}$ and $\mathcal{L}_{as}$. Interestingly, the case without $\mathcal{L}_b$ (Eq.(12)) receives the minimal accuracy loss (4.29% and 0.16%), confirming the importance of $\mathcal{L}_b$ on the basis of $\mathcal{L}_{ds}$ and $\mathcal{L}_{as}$.

Visualization of Generated Samples

To further show the desirable adaptability of the generated samples by AdaSG to Q, we conduct the visual analysis over MobileNetV2 serving as both P and Q on ImageNet by the similarity matrix (each element is obtained by computing the ℓ_1 norm between the probability distribution p_{ds} of every two samples), along with the visualization of generated samples in Fig.7. Fig.7(a) illustrates that the generated samples by AdaSG possess a much larger similarity (the darker, the larger) than those by GDFQ, implying that the generated sample by GDFQ varies greatly, where a lots of samples with undesirable adaptability exist against AdaSG. Fig.7(b) shows that the samples generated for 3-bit and 5-bit precision vary greatly, while the samples from varied categories also differ greatly from each other, verifying that the samples possess desirable adaptability to varied Q, upon the fact that the category (label) information is fully exploited.

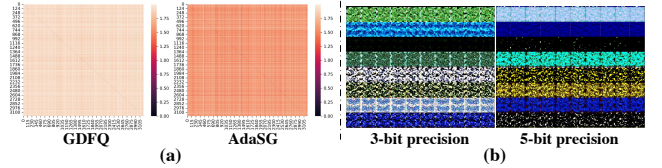


Figure 7: (a) The similarity comparison between the generated samples. (b) Visualization of the generated samples, where each row denotes one of 8 randomly chosen classes from ImageNet.

Conclusion

In this paper, we rethink date-free quantization process as a zero-sum game between two players — a generator and a quantized network, then further develop an Adaptability-aware Sample Generation (AdaSG) method, which features a dynamic maximization-vs-minimization game process anchored on sample adaptability. The maximization process generates the sample with desirable adaptability, which is further reduced by the minimization process after recovering the performance of Q. The Balance Gap is defined to achieve the stationarity for the zero-sum game. The theoretical analysis and empirical studies validate the advantages of AdaSG to the existing arts.

Acknowledgments

This work are supported by National Natural Science Foundation of China under the grant no U21A20470, 62172136, 72188101, U1936217. Key Research and Technology Development Projects of Anhui Province (no.202004a5020043).

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